

Entity Analysis Report

Project Title: E-Commerce Order Management Database System

Phase: Week 2 - Entity and Relationship Analysis

1. Entity Identification

Based on standard requirements for an E-Commerce Order Management system, the following core entities have been identified:

- **Customer:** Stores details of users who place orders.
- **Category:** Classifies products into logical groups.
- **Product:** Stores details of items available for purchase.
- **Order:** Tracks the overarching order placed by a customer.
- **OrderItem:** Details individual products within a specific order (resolves the many-to-many relationship between Orders and Products).
- **Payment:** Records transactions associated with orders.

2. Entity Attribute List & Constraints

This section outlines the attributes, Primary Keys (PK), Foreign Keys (FK), and constraints (NOT NULL, UNIQUE, CHECK, DEFAULT) for each entity.

Customer Entity

Attribute Name	Data Type	Key	Constraints
CustomerID	INT	PK	NOT NULL, AUTO_INCREMENT
FirstName	VARCHAR(50)	-	NOT NULL
LastName	VARCHAR(50)	-	NOT NULL
Email	VARCHAR(100)	-	NOT NULL, UNIQUE
Phone	VARCHAR(15)	-	UNIQUE
RegistrationDate	DATETIME	-	DEFAULT CURRENT_TIMESTAMP

Category Entity

Attribute Name	Data Type	Key	Constraints
CategoryID	INT	PK	NOT NULL, AUTO_INCREMENT
CategoryName	VARCHAR(100)	-	NOT NULL, UNIQUE
Description	TEXT	-	-

Product Entity

Attribute Name	Data Type	Key	Constraints
ProductID	INT	PK	NOT NULL, AUTO_INCREMENT
CategoryID	INT	FK	NOT NULL
ProductName	VARCHAR(150)	-	NOT NULL
Price	DECIMAL(10,2)	-	NOT NULL, CHECK (Price > 0)
StockQuantity	INT	-	NOT NULL, CHECK (StockQuantity >= 0), DEFAULT 0

Order Entity

Attribute Name	Data Type	Key	Constraints
OrderID	INT	PK	NOT NULL, AUTO_INCREMENT
CustomerID	INT	FK	NOT NULL
OrderDate	DATETIME	-	DEFAULT CURRENT_TIMESTAMP
TotalAmount	DECIMAL(10,2)	-	NOT NULL, CHECK (TotalAmount >= 0)
Status	VARCHAR(20)	-	DEFAULT 'Pending', CHECK (Status IN ('Pending', 'Processing', 'Shipped', 'Delivered', 'Cancelled'))

OrderItem Entity

Attribute Name	Data Type	Key	Constraints
OrderItemID	INT	PK	NOT NULL, AUTO_INCREMENT
OrderID	INT	FK	NOT NULL

ProductID	INT	FK	NOT NULL
Quantity	INT	-	NOT NULL, CHECK (Quantity > 0)
UnitPrice	DECIMAL(10,2)	-	NOT NULL, CHECK (UnitPrice >= 0)

Payment Entity

Attribute Name	Data Type	Key	Constraints
PaymentID	INT	PK	NOT NULL, AUTO_INCREMENT
OrderID	INT	FK	NOT NULL, UNIQUE
PaymentDate	DATETIME	-	DEFAULT CURRENT_TIMESTAMP
Amount	DECIMAL(10,2)	-	NOT NULL, CHECK (Amount > 0)
PaymentMethod	VARCHAR(50)	-	NOT NULL

3. Relationship & Cardinality Analysis

Customer & Order

Relationship: A Customer can place multiple Orders, but each Order is placed by exactly one Customer.

Cardinality: **One-to-Many (1:N)**

Category & Product

Relationship: A Category can contain multiple Products, but each Product belongs to exactly one Category.

Cardinality: **One-to-Many (1:N)**

Order & OrderItem

Relationship: An Order can contain multiple OrderItems (different products), but each OrderItem belongs to exactly one Order.

Cardinality: **One-to-Many (1:N)**

Product & OrderItem

Relationship: A Product can appear in multiple OrderItems (across different orders), but each OrderItem references exactly one Product.

Cardinality: **One-to-Many (1:N)**

Order & Payment

Relationship: An Order has one associated Payment transaction, and a Payment is made for exactly one Order.

Cardinality: **One-to-One (1:1)**